

Table of contents

\\Research

Investigators

XChen

31P CSI 20230503 NOE

localizer AAHead_Scout T1_MEMPRAGE_64ch fastestmap fastestmap fastestmap fastestmap SPECIAL_ACC head_csi_fid head_csi_fid Ax TSE PD T2 t2_tse_dark-fluid_tra resolve_4scan_trace_tra_p2_192
--

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\localizer

TA: 0:19 PM: REF Voxel size: 0.5×0.5×10.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	5
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
Filter	Normalize, Elliptical filter
Coil elements	1H

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0 ms
MTC	Off
Magn. preparation	None
Flip angle	40 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Contrast - Dynamic

Averages	1
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1

Contrast - Dynamic

Multiple series	Each measurement
-----------------	------------------

Resolution - Common

FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
Base resolution	256
Phase resolution	50 %
Phase partial Fourier	Off
Interpolation	On

Resolution - iPAT

PAT mode	None
----------	------

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	On
Unfiltered images	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	On

Geometry - Common

Slice group	1
Slices	5
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
TR	20.0 ms
Multi-slice mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice group	1
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Position	Isocenter

Geometry - AutoAlign

Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Rotation	0.00 deg
Initial Orientation	Sagittal

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Tune up
B1 Shim mode	TrueForm
Adjust with body coil	true
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	20.0 ms
Concatenations	7
Segments	1

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	280 mm
FoV phase	100.0 %
Phase resolution	50 %

Physio - PACE

Resp. control	Off
Concatenations	7

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Sequential
Bandwidth	180 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Fast
Gradient mode	Normal
Excitation	Slice-sel.
RF spoiling	On

Sequence - Assistant

Mode	Off
Allowed delay	0 s

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\AAHead_Scout

TA: 0:14 PM: REF Voxel size: 1.6×1.6×1.6 mmPAT: 3 Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	On
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slab group	1
Slabs	1
Dist. factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Phase oversampling	0 %
Slice oversampling	0.0 %
Slices per slab	128
FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
Filter	Prescan Normalize
Coil elements	HE1-4

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip angle	8 deg

Contrast - Dynamic

Averages	1
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1

Resolution - Common

FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
Base resolution	160
Phase resolution	100 %
Slice resolution	69 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Trajectory	Cartesian

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	3
Ref. lines PE	24
Accel. factor 3D	1

Resolution - iPAT

Reference scan mode	Integrated
---------------------	------------

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slab group	1
Slabs	1
Dist. factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice oversampling	0.0 %
Slices per slab	128
FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
TR	3.15 ms
Multi-slice mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Rotation	0.00 deg
Initial Orientation	Transversal

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Tune up
B1 Shim mode	TrueForm
Adjust with body coil	Off

System - Adjustments

Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

Sequence - Part 2

RF spoiling	On
-------------	----

Sequence - Assistant

Mode	Off
------	-----

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	Low
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Flip angle	8 deg
Measurements	1
Time to center	6.2 s

Inline - Inline

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	On
Dimension	3D
Asymmetric echo	Weak
Contrasts	1
Multi-slice mode	Sequential
Bandwidth	540 Hz/Px

Sequence - Part 2

RF pulse type	Fast
Gradient mode	Normal
Excitation	Non-sel.

\\Research\Investigators\XChen\31P CSI 20230503 NOE\T1_MEMPRAGE_64ch

TA: 6:30 PM: REF Voxel size: 1.0×1.0×1.0 mmPAT: Off Rel. SNR: 1.00 : tfl_me

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	On
Start measurements	Single measurement

Routine

Slab group	1
Slabs	1
Dist. factor	50 %
Position	L6.8 A11.3 F21.5 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
Slice oversampling	0.0 %
Slices per slab	176
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
Filter	Raw filter, Normalize
Coil elements	1H

Contrast - Common

TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. preparation	Non-sel. IR
TI	1100 ms
Flip angle	7.0 deg
Fat suppr.	None
Water suppr.	None

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	1.00 mm

Resolution - Common

Base resolution	256
Phase resolution	60 %
Slice resolution	100 %
Phase partial Fourier	Off
Slice partial Fourier	Off
Interpolation	Off

Resolution - iPAT

PAT mode	None
----------	------

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	On
Unfiltered images	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	On
Elliptical filter	Off

Geometry - Common

Slab group	1
Slabs	1
Dist. factor	50 %
Position	L6.8 A11.3 F21.5 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice oversampling	0.0 %
Slices per slab	176
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	2530.0 ms
Multi-slice mode	Single shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab group	1
Position	L6.8 A11.3 F21.5 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L6.8 A11.3 F21.5
L	6.8 mm
A	11.3 mm
F	21.5 mm
Initial Rotation	0.01 deg
Initial Orientation	Sagittal

Geometry - Navigator**System - Miscellaneous**

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L

System - Miscellaneous

Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	true
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L6.8 A11.3 F21.5 mm
Orientation	Sagittal
Rotation	0.01 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	Low
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2530.0 ms
Concatenations	1

Physio - Cardiac

Magn. preparation	Non-sel. IR
TI	1100 ms
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	60 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reordering	Linear
Asymmetric echo	Off
Contrasts	4
Flow comp. 1	No
Multi-slice mode	Single shot
Echo spacing	9.8 ms
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px

Sequence - Part 2

RF pulse type	Fast
Gradient mode	Fast
Excitation	Non-sel.
RF spoiling	On
Incr. Gradient spoiling	Off
Turbo factor	176

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1
Averaging	RMS

Sequence - Assistant

Mode	Off
------	-----

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\fastestmap

TA: 8.5 s PM: FIX Vol: 50 ×28 ×50 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800 ms
TE	50.00 ms
Averages	1
Filter	None
Coil elements	1H

Contrast

TR	2800 ms
TE	50.00 ms
Tau	5.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256
-------------	-----

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol R >> L	50 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	L7.4 A35.1 H17.8
L	7.4 mm
A	35.1 mm
H	17.8 mm
Initial Rotation	90.00 deg
Initial Orientation	C > T
C > T	-17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H

System - Miscellaneous

Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2800 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Linear 3-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm

Sequence - Special

Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\Investigators\XChen\31P CSI 20230503 NOE\fastestmap

TA: 0:17 PM: FIX Vol: 50 ×28 ×50 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800 ms
TE	50.00 ms
Averages	1
Filter	None
Coil elements	1H

Contrast

TR	2800 ms
TE	50.00 ms
Tau	5.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256
-------------	-----

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol R >> L	50 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	L7.4 A35.1 H17.8
L	7.4 mm
A	35.1 mm
H	17.8 mm
Initial Rotation	90.00 deg
Initial Orientation	C > T
C > T	-17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H

System - Miscellaneous

Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	true
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2800 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm

Sequence - Special

Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\fastestmap

TA: 0:17 PM: FIX Vol: 50 ×28 ×50 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800 ms
TE	50.00 ms
Averages	1
Filter	None
Coil elements	1H

Contrast

TR	2800 ms
TE	50.00 ms
Tau	5.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256
-------------	-----

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol R >> L	50 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	L7.4 A35.1 H17.8
L	7.4 mm
A	35.1 mm
H	17.8 mm
Initial Rotation	90.00 deg
Initial Orientation	C > T
C > T	-17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H

System - Miscellaneous

Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	true
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2800 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm

Sequence - Special

Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\Investigators\XChen\31P CSI 20230503 NOE\fastestmap

TA: 0:34 PM: FIX Vol: 50 ×28 ×50 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800 ms
TE	50.00 ms
Averages	1
Filter	None
Coil elements	1H

Contrast

TR	2800 ms
TE	50.00 ms
Tau	15.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256
-------------	-----

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0 deg
Vol F >> H	28 mm
Vol R >> L	50 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	L7.4 A35.1 H17.8
L	7.4 mm
A	35.1 mm
H	17.8 mm
Initial Rotation	90.00 deg
Initial Orientation	C > T
C > T	-17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H

System - Miscellaneous

Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2800 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Linear 6-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm

Sequence - Special

Inversion pulse	Off
Multi-echo acquisition	Off

\\Research\Investigators\XChen\31P CSI 20230503 NOE\SPECIAL_ACC

TA: 1:48 PM: FIX Vol: 32 ×20 ×34 mmRel. SNR: 1.00 : svs_se

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

Routine

Position	L2.3 A45.1 H61.3 mm
Orientation	C > T-24.6 > S4.6
Rotation	1.262618 deg
Vol F >> H	20 mm
Vol F >> H	20 mm
Vol A >> P	34 mm
TR	3000 ms
TE	12.00 ms
Averages	32
Filter	None
Coil elements	HC3-6

System - Adjustments

B0 Shim mode	Advanced
B1 Shim mode	TrueForm
Adj. water suppr.	On
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

Contrast

TR	3000 ms
TE	12.00 ms
Averages	32
Flip angle	90 deg
Water suppr.	Water sat.
Water suppr. BW	45 Hz
Spectral suppr.	None
Measurements	1

System - Adjust Volume

Position	L2.3 A45.1 H61.3 mm
Orientation	C > T-24.6 > S4.6
Rotation	91.26 deg
F >> H	20 mm
R >> L	32 mm
A >> P	34 mm
Reset	Off

Resolution - Common

Prescan Normalize	Off
Vector size	2048

Geometry - Common

Position	L2.3 A45.1 H61.3 mm
Orientation	C > T-24.6 > S4.6
Rotation	1.262618 deg
Vol F >> H	20 mm
Vol R >> L	32 mm
Vol A >> P	34 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	L2.3 A45.1 H61.3
L	2.3 mm
A	45.1 mm
H	61.3 mm
Initial Rotation	91.26 deg
Initial Orientation	C > T
C > T	-24.6
> S	4.6

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Physio - PACE

Resp. control	Off
---------------	-----

Sequence - Common

Preparation scans	4
Delta frequency	-2.3 ppm
Ref. scan mode	Off
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	1024 ms
Remove oversampling	On

Sequence - Special

Inversion Pulse Duration	9000 us
--------------------------	---------

Geometry - Navigator

Sequence - Special

Adiabatic Inversion	On
Inversion Delay	25000 us
Excitation Pulse Duration	3200 us
Refoc Pulse Duration	8000 us
VAPOR water suppression	On
VAPOR Flip Angle	110 degrees
Final VAPOR Delay	18000 us
Additional water suppression	On

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\head_csi_fid

TA: 26:36 PM: FIX Voxel size: 50.0×50.0×36.0 mmRel. SNR: 1.00 : csi_fid

Properties

Prio recon	Off
Load images to viewer	Off
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0 deg
Slices	1
Vol A >> P	400 mm
Vol R >> L	400 mm
FoV A >> P	400 mm
FoV R >> L	400 mm
Thickness F >> H	36 mm
TR	3000 ms
TE	2.3 ms
Averages	60
Filter	None
Coil elements	31P

Contrast

TR	3000 ms
TE	2.3 ms
Averages	60
Averaging mode	Long term
Flip angle	90 deg
Water suppr.	None
Measurements	1

Resolution - Common

FoV R >> L	400 mm
FoV A >> P	400 mm
Thickness F >> H	36 mm
Scan res. R >> L	8
Scan res. A >> P	8
Interpol. res. R >> L	8
Interpol. res. A >> P	8
Hamming	Off
Prescan Normalize	Off
Vector size	512

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0 deg
FoV R >> L	400 mm
FoV A >> P	400 mm
Thickness F >> H	36 mm
Vol R >> L	400 mm
Vol A >> P	400 mm

Geometry - AutoAlign

Slice group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Phase enc. dir.	A >> P
AutoAlign	---
Initial Position	L30.0 A10.7 H24.3
L	30.0 mm
A	10.7 mm
H	24.3 mm
Initial Rotation	0.00 deg
Initial Orientation	T > C
T > C	17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Advanced
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	36 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 31P	49.891914 MHz
Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 31P	0.000 V
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
-----------------	------

Physio - Signal1

TR	3000 ms
----	---------

Sequence - Common

Preparation scans	8
Dimension	2D
Delta frequency	0.0 ppm
Phase encoding	Weighted
Acquisition delay	2.3 ms
Bandwidth	4000 Hz
Acquisition duration	128 ms
Remove oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX delta frequency	0 Hz
NOE type	Rectangular
NOE count	1
NOE duration	5.0 ms
NOE flip angle	90 deg
TX Nucleus	1H
TX delta frequency	0 Hz
Decoupling type	Waltz4
Decoupling duration	1.00 ms
DC total duration	50 %
DC pause fract.	100 %
Decoupling flip angle	180 deg
Coil elements	31P

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\head_csi_fid

TA: 10:30 PM: FIX Voxel size: 50.0×50.0×40.0 mmRel. SNR: 1.00 : csi_fid

Properties

Prio recon	Off
Load images to viewer	Off
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0 deg
Slices	1
Vol A >> P	400 mm
Vol R >> L	400 mm
FoV A >> P	400 mm
FoV R >> L	400 mm
Thickness F >> H	40 mm
TR	3000 ms
TE	2.3 ms
Averages	22
Filter	Hamming
Coil elements	31P

Contrast

TR	3000 ms
TE	2.3 ms
Averages	22
Averaging mode	Short term
Flip angle	90 deg
Water suppr.	None
Measurements	1

Resolution - Common

FoV R >> L	400 mm
FoV A >> P	400 mm
Thickness F >> H	40 mm
Scan res. R >> L	8
Scan res. A >> P	8
Interpol. res. R >> L	8
Interpol. res. A >> P	8
Hamming	On
Width	50
Prescan Normalize	Off
Vector size	512

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0 deg
FoV R >> L	400 mm
FoV A >> P	400 mm
Thickness F >> H	40 mm
Vol R >> L	400 mm
Vol A >> P	400 mm

Geometry - AutoAlign

Slice group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Phase enc. dir.	A >> P
AutoAlign	---
Initial Position	L30.0 A10.7 H24.3
L	30.0 mm
A	10.7 mm
H	24.3 mm
Initial Rotation	0.00 deg
Initial Orientation	T > C
T > C	17.0
> S	0.0

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Advanced
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	40 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 31P	49.891914 MHz
Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 31P	0.000 V
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
-----------------	------

Physio - Signal1

TR	3000 ms
----	---------

Sequence - Common

Preparation scans	8
Dimension	2D
Delta frequency	0.0 ppm
Phase encoding	Weighted
Acquisition delay	2.3 ms
Bandwidth	4000 Hz
Acquisition duration	128 ms
Remove oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX delta frequency	0 Hz
NOE type	Rectangular
NOE count	10
NOE duration	5.0 ms
NOE pause	10.0 ms
NOE flip angle	90 deg
TX Nucleus	1H
TX delta frequency	0 Hz
Decoupling type	Waltz4
Decoupling duration	1.00 ms
DC total duration	50 %
DC pause fract.	100 %
Decoupling flip angle	180 deg
Coil elements	31P

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\Ax TSE PD T2

TA: 3:32 PM: REF Voxel size: 0.4×0.4×4.0 mmPAT: Off Rel. SNR: 1.00 : tse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
TR	3000.0 ms
TE 1	9 ms
TE 2	81 ms
Averages	1
Concatenations	2
Filter	Prescan Normalize
Coil elements	HE1-4

Contrast - Common

TR	3000.0 ms
TE 1	9 ms
TE 2	81 ms
TD	0.0 ms
MTC	Off
Magn. preparation	None
Flip angle	120 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	Off

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
Base resolution	256
Phase resolution	80 %
Phase partial Fourier	Off
Trajectory	Cartesian

Resolution - Common

Interpolation	On
---------------	----

Resolution - iPAT

PAT mode	None
----------	------

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
TR	3000.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Rotation	90.00 deg
Initial Orientation	Transversal

Geometry - Saturation

Fat suppr.	None
Water suppr.	None
Restore magn.	Off
Special sat.	None

Geometry - Navigator**System - Miscellaneous**

Positioning mode	REF
Table position	H
Table position	37 mm
MSMA	S - C - T
Sagittal	R >> L

System - Miscellaneous

Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	171 mm
A >> P	210 mm
F >> H	149 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000.0 ms
Concatenations	2

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	210 mm
FoV phase	81.3 %
Phase resolution	80 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	2

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
---------	-----

Inline - MIP

MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	On
Contrasts	2
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	8.96 ms
Bandwidth	250 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	34
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Normal
Hyperecho	Off
WARP	Off
Red. EC sensitivity	Off
Turbo factor	5

Sequence - Assistant

Mode	Min flip angle
Min flip angle	130 deg
Allowed delay	60 s

\\Research\Investigators\XChen\31P CSI 20230503 NOE\t2_tse_dark-fluid_tra

TA: 4:50 PM: FIX Voxel size: 0.8×0.8×4.0 mmPAT: Off Rel. SNR: 1.00 : tir

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
TR	9000.0 ms
TE	114 ms
Averages	1
Concatenations	2
Filter	Prescan Normalize, Elliptical filter
Coil elements	HE1-4

Contrast - Common

TR	9000.0 ms
TE	114 ms
TD	0.0 ms
MTC	Off
Magn. preparation	Slice-sel. IR
TI	2500 ms
Flip angle	130 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Freeze suppressed tissue	Off

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
Base resolution	256
Phase resolution	90 %

Resolution - Common

Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off

Resolution - iPAT

PAT mode	None
----------	------

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	On

Geometry - Common

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
FoV read	210 mm
FoV phase	81.3 %
Slice thickness	4.0 mm
TR	9000.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Rotation	90.00 deg
Initial Orientation	Transversal

Geometry - Saturation

Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Special sat.	None

Geometry - Navigator**System - Miscellaneous**

Positioning mode	FIX
Table position	H

System - Miscellaneous

Table position	37 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	171 mm
A >> P	210 mm
F >> H	149 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	9000.0 ms
Concatenations	2

Physio - Cardiac

Magn. preparation	Slice-sel. IR
TI	2500 ms
Fat suppr.	Fat sat.
Dark blood	Off
FoV read	210 mm
FoV phase	81.3 %
Phase resolution	90 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	2

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off

Inline - Common

Save original images	On
----------------------	----

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	10.3 ms
Bandwidth	179 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	15
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Hyperecho	Off
WARP	Off
Red. EC sensitivity	Off
Turbo factor	13

Sequence - Assistant

Mode	Off
Allowed delay	60 s

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\resolve_4scan_trace_tra_p2_192

TA: 2:34 PM: FIX Voxel size: 1.1×1.1×4.0 mmPAT: Off Rel. SNR: 1.00 : resolve

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	217 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	5860 ms
TE 1	83 ms
TE 2	152 ms
Concatenations	1
Filter	Raw filter, Prescan Normalize
Coil elements	HE1-4

Contrast - Common

TR	5860 ms
TE 1	83 ms
TE 2	152 ms
MTC	Off
Magn. preparation	None
Flip angle	180 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong

Contrast - Dynamic

Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1

Resolution - Common

FoV read	217 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
Base resolution	192
Phase resolution	100 %
Phase partial Fourier	Off
Readout partial Fourier	5/8
Readout segments	7
Interpolation	Off

Resolution - iPAT

PAT mode	None
----------	------

Resolution - Filter Image

Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off

Resolution - Filter Rawdata

Raw filter	On
------------	----

Geometry - Common

Slice group	1
Slices	30
Dist. factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	A >> P
FoV read	217 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	5860 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Rotation	0.00 deg
Initial Orientation	Transversal

Geometry - Saturation

Fat suppr.	Fat sat.
Fat sat. mode	Strong
Special sat.	None

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	37 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm

System - Adjustments

Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	217 mm
R >> L	217 mm
F >> H	149 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
--------------	----------

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	5860 ms
Concatenations	1

Diff - Neuro

Diffusion mode	4-Scan Trace
Diff. directions	4
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
b-value 1	1
b-value 2	1
Diff. weighted images	Off
Trace weighted images	On
ADC maps	On
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	100

Diff - Body

Diffusion mode	4-Scan Trace
Diff. directions	4
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
b-value 1	1
b-value 2	1
Diff. weighted images	Off
Trace weighted images	On
ADC maps	On
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off

Diff - Body

b-Value >=	0 s/mm ²
Noise level	100

Diff - Composing

Distortion Corr.	Off
------------------	-----

Sequence - Part 1

Introduction	On
Dimension	2D
Contrasts	2
Optimization	Min. TE
Multi-slice mode	Interleaved
Echo spacing	0.32 ms
Bandwidth	723 Hz/Px

Sequence - Part 2

EPI factor	192
RF pulse type	Normal
Gradient mode	Fast
Reacquisition mode	On

Sequence - Assistant

Mode	Off
------	-----